Math 50 Final: Practice Problems

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Exercise 1: Consider the following code.

```
> n = 100000
> x1 = np.random.normal(0, 2, n)
> x2 = 0.5*x1 + np.random.normal(0, 1, n)
> y = 2*x1 + 3*x2 + np.random.normal(0, 1, n)
```

(a) What is the slope β_1' when Y is regressed on X_1 alone?

(b) What is the slope β_2' when Y is regressed on X_2 alone?

Exercise 2: Consider the following code.

```
> n = 100000
> x = np.random.choice([0, 1, 2], size=n, p=[0.5, 0.3, 0.2])
> y = np.random.normal(1 + 2*x, 1)
> print(np.mean(y))
> print(np.var(y))
> print(np.mean(y[x == 2]))
```

(a) Write the model being simulated in $\sim$ notation, for X and $Y \mid X$.

(b) What (approximately) do the three `print` statements produce?

Exercise 3: Consider the single-predictor model $Y = \beta_1 X + \epsilon$ with $E[X] = 0$ and no intercept, fit to N points with $\hat{\sigma}_X^2 = \frac{1}{N}\sum_{i=1}^{N} X_i^2$. Recall $\text{se}(\hat{\beta}_1) = \sigma_\epsilon^2/(N\hat{\sigma}_X^2)\big)$. A regularized estimator of β_1 is

$$\hat{\beta}_1^\lambda = \frac{\hat{\beta}_1}{1 + \lambda/\hat{\sigma}_X^2}.$$

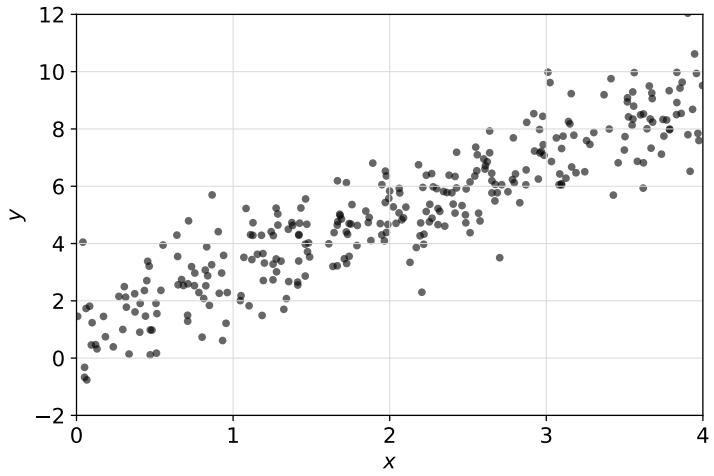
Suppose the true slope is $\beta_1 = 3$. Give your answers in terms of λ, $\hat{\sigma}_X^2$, σ_ϵ^2 and N.

(a) Compute the bias and the variance of $\hat{\beta}_1^\lambda$.

(b) Compute the MSE of $\hat{\beta}_1^\lambda$.

(c) Compute $\mathrm{MSE}_{\hat{y}(2,D)} = E_D\big[(\hat{y}(2, D) - f(2))^2\big]$.

Exercise 4: The plot shows 300 points from a model $Y \mid X \sim \mathrm{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$.



(a) Estimate β_0 and β_1 from the plot.

(b) Estimate σ_ϵ. Explain how you read it off the plot.

(c) Estimate $\mathrm{var}(X)$.

Exercise 5: Consider the model

$$Y = \sum_{j=1}^{K} \beta_j \phi_j(X) + \epsilon,$$

with no intercept, fit by least squares to N points (X_i, Y_i). Treat the X_i as fixed, and suppose the noise terms ϵ_i are independent with variance σ_ϵ^2. The features are uncorrelated and mean zero in the data:

$$\frac{1}{N}\sum_{i=1}^{N} \phi_j(X_i)\phi_k(X_i) = 0 \quad (j \neq k), \qquad \frac{1}{N}\sum_{i=1}^{N} \phi_j(X_i) = 0.$$

(a) Write the least-squares estimate $\hat{\beta}_j$ in terms of the data, and compute $\mathrm{var}(\hat{\beta}_j)$.

(b) Show that $\mathrm{cov}(\hat{\beta}_j, \hat{\beta}_k) = 0$ for $j \neq k$.

(c) The fit is $\hat{y}(x, D) = \sum_{j=1}^{K} \hat{\beta}_j \phi_j(x)$. Show that the variance of the fit, averaged over the data points,

$$\frac{1}{N}\sum_{i=1}^{N} \mathrm{var}\big(\hat{y}(X_i, D)\big),$$

equals $K\sigma_\epsilon^2/N$, so it grows linearly in the number of features K.

Exercise 6: In a class, $X = 1$ if a student did the practice problems (0 otherwise) and $Y = 1$ if the student passed the exam (0 otherwise). The joint distribution is

	$Y = 0$	$Y = 1$
$X = 0$	0.4	0.1
$X = 1$	0.2	0.3

(a) Compute $E\left[X^{50}Y^{7}\right]$.

(b) Compute $E\left[(X+Y)^{2}\right]$.

(c) Compute $\text{var}\left(X^{10}Y^{10}\right)$ and $\text{cov}(X, Y)$.

Exercise 7: We can view a political system as a way of *estimating*, from citizens' votes, which policies best satisfy some set of criteria. Consider three cases:

- **Direct democracy:** each issue is decided by a popular vote.
- **Representative democracy:** elected officials decide many issues on voters' behalf.
- **Dictatorship:** a small number of individuals make all the decisions.

Discuss the pros and cons of these systems in terms of the bias-variance tradeoff.

Exercise 8: A utility company models the monthly electricity use Y (in kWh) of homes using the floor area X_{area} (in square feet) and the number of rooms X_{rooms}. In their data the two predictors have correlation 0.95. They fit

$$Y = \beta_0 + \beta_{\text{area}} X_{\text{area}} + \beta_{\text{rooms}} X_{\text{rooms}} + \epsilon$$

and find $R^2 = 0.85$, but the 95% confidence intervals for both β_{area} and β_{rooms} contain 0. When they fit a model with X_{area} alone, the confidence interval for its slope is far from 0.

(a) Does the two-predictor fit suggest that floor area has no effect on electricity use? Explain in words; you do not need to do a calculation.

(b) What does this dataset determine well, and what would you need to separate the effect of floor area from the effect of the number of rooms?